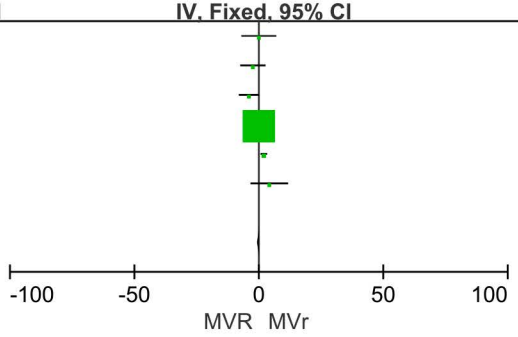
A

Study or Subgroup	MVR Mean	MVR SD	MVR Total	MVr Mean	MVr SD	MVr Total	Weight	Mean Difference IV, Fixed, 95% CI
DiGregorio 2004	63	12	13	63	9	46	0.1%	0.00 [-7.02, 7.02]
Gonçalo 2015	61.1	13	26	63.5	10.9	475	0.1%	-2.40 [-7.49, 2.69]
Helder 2014	56	8	21	60	7	40	0.2%	-4.00 [-8.05, 0.05]
Hendrix 2019	60	5	3520	60	5	8523	97.3%	0.00 [-0.20, 0.20]
Lazam 2017	66	9	213	64	10	1709	2.2%	2.00 [0.70, 3.30]
Pandis 2011	62.3	13.3	19	58.1	10.8	21	0.1%	4.20 [-3.36, 11.76]
Total (95% CI)			**3812**			**10814**	**100.0%**	**0.03 [-0.16, 0.23]**

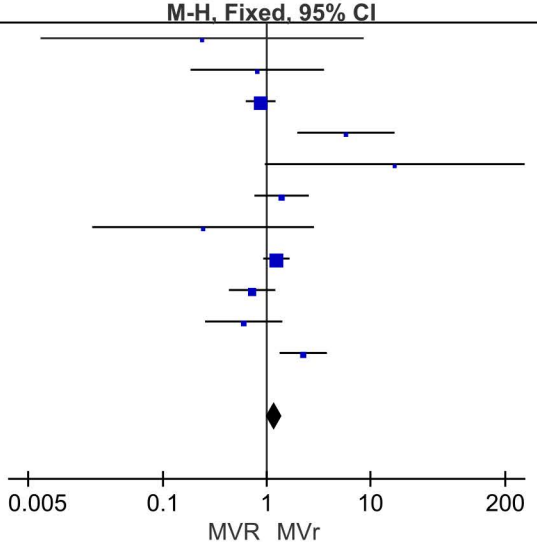
Heterogeneity: Chi² = 14.78, df = 5 (P = 0.01); I² = 66%
Test for overall effect: Z = 0.35 (P = 0.73)



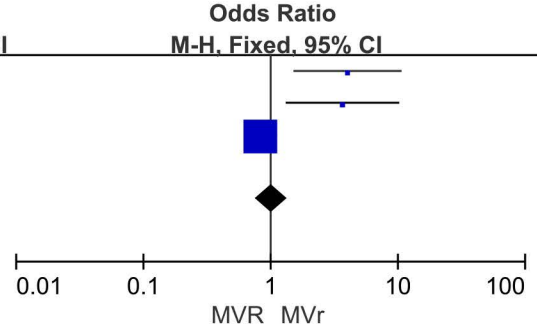
B

Study or Subgroup	MVR Events	MVR Total	MVr Events	MVr Total	Weight	Odds Ratio M-H, Fixed, 95% CI
Bozbuga 2003	2	3	3	3	0.5%	0.24 [0.01, 8.62]
DiGregorio 2004	10	13	37	46	1.5%	0.81 [0.18, 3.57]
Gillinov 2003	81	232	170	447	30.4%	0.87 [0.63, 1.22]
Gonçalo 2015	22	26	231	475	1.5%	5.81 [1.97, 17.11]
Gramalia 1999	63	63	55	62	0.2%	17.16 [0.96, 307.35]
Hata 2019	46	85	39	85	7.2%	1.39 [0.76, 2.54]
Helder 2014	19	21	39	40	1.0%	0.24 [0.02, 2.86]
Lazam 2017	83	213	581	1709	31.6%	1.24 [0.92, 1.66]
Lee 1997	72	111	120	167	13.5%	0.72 [0.43, 1.21]
Zalaquett 2005	14	28	55	88	5.3%	0.60 [0.25, 1.41]
Zhou 2010	37	78	69	241	7.1%	2.25 [1.33, 3.80]
Total (95% CI)		**873**		**3363**	**100.0%**	**1.18 [1.00, 1.40]**
Total events	449		1399			

Heterogeneity: Chi² = 29.49, df = 10 (P = 0.001); I² = 66%
Test for overall effect: Z = 1.94 (P = 0.05)



C

Study or Subgroup	MVR Events	MVR Total	MVr Events	MVr Total	Weight	Odds Ratio M-H, Fixed, 95% CI
Gonçalo 2015	6	26	33	475	2.7%	4.02 [1.51, 10.69]
Shafii 2012	15	30	9	42	3.9%	3.67 [1.31, 10.24]
Vassileva 2011	348	1036	83	219	93.4%	0.83 [0.61, 1.12]
Total (95% CI)		**1092**		**736**	**100.0%**	**1.02 [0.77, 1.35]**
Total events	369		125			

Heterogeneity: Chi² = 15.31, df = 2 (P = 0.0005); I² = 87%
Test for overall effect: Z = 0.17 (P = 0.87)